

Exploring Influences of Design and Environmental Factors on Recognizing Teacher's Facial Expressions in Educational VR

Yu Han^{*†}Jie Hao^{*}

Yu Miao

Yue Liu[‡] (✉)

Beijing Engineering Research Center of Mixed Reality and Advanced Display
School of Optics and Photonics, Beijing Institute of Technology

ABSTRACT

Teachers' facial expressions significantly influence students' willingness to learn. However, insufficient resolution of current consumer head-mounted displays leads to difficulties in capturing nuances of virtual teachers in VR, which may be solved by scaling up the heads of virtual teachers. In this paper, we explore the effects of design and environmental factors on the ideal head scales for recognizing the virtual teacher's facial expressions. Our results show that the facial visualization style and the distance from the user to the virtual teacher are important factors affecting facial expression recognition, which could contribute to the design and optimization of educational VR.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI; Applied computing—Education—Interactive learning environments

1 INTRODUCTION

Virtual reality (VR) holds great potential for new didactic concepts in teaching, since learning environments can be generated digitally. Virtual teachers are widely used to provide vivid teacher-student interaction and increase the learning effectiveness of learners through facial expressions and body gestures in educational VR. Previous studies have suggested that teachers' facial expressions play an important role in students' learning process [4, 5]. For example, teachers' eye contact and facial expressions positively influence students' willingness to attend to and participate in class [4]. However, currently, the resolution of consumer head-mounted displays (HMDs) is insufficient, which becomes an obstacle in catching every nuance of virtual teachers' facial expressions.

To deal with such a problem, some research efforts have been conducted to scale up the head of a virtual human, which enables users to easily distinguish his/her facial expressions [1, 2]. However, the impacts of design and environmental factors on recognizing the facial expressions of virtual humans have not been thoroughly investigated, which limits the usability and user experience of such an approach in educational VR.

In this paper, we choose three factors, namely the surrounding context (i.e. neutral and classroom-style), the facial visualization style (i.e. cartoon and realistic) and the distance from the user to the virtual teacher (i.e. 1m, 3m, and 6m), and explore their influence on the ideal head scales for facial-expression-recognizing tasks. This research could contribute to the design and optimization of educational VR where virtual teachers are used to give lectures.

2 METHODOLOGY

In this study, a $2 \times 2 \times 3$ factorial within-subjects design was adopted given three factors: the surrounding context where the virtual teacher and the user were located, the facial visualization style of the virtual teacher, and the distance from the user to the virtual teacher. The levels of the surrounding context were neutral and classroom-style; the levels of the facial visualization style were cartoon and realistic; and the levels of the distance were 1 m, 3 m and 6 m.

2.1 Participants

A total of 19 participants (10 males and 9 females, ages ranging from 21 to 31, $M = 24.4$, $SD = 2.4$), recruited from a university, took part in this study.

2.2 Apparatus

The experimental setup includes an HTC Vive Pro with a Vive controller and two Lighthouse units, a 24-inch AOC monitor, and a host PC with Microsoft Windows 10. Unity engine (version 2020.3.0.f1) was employed for implementing VR techniques.

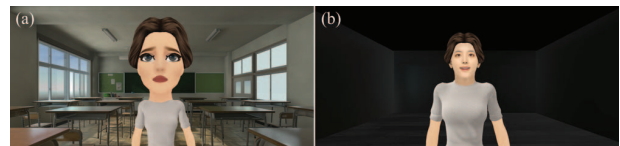


Figure 1: Examples of different head scales of virtual teachers with different facial visualization styles and surrounding contexts: (a) big head (2x) with cartoon style in a classroom-style context; (b) normal head (1x) with realistic style in a neutral context.

2.3 Experimental Design

An experimental system named "Big Head" was designed to implement the experiment. We prepared two virtual contexts (i.e. neutral and classroom-style) and two life-size 3D virtual female teachers with different facial visualization styles (i.e. cartoon and realistic), as shown in Fig. 1. In the experiment, each virtual teacher will be located at three different positions in each virtual context, which means that the teacher is located at 1 m, 3 m and 6 m away from the participants, respectively. The selection of distances is based on Hall's Proxemics theory [3] and the actual reality of class. This generates 12 combinations in total. The size of the virtual classroom is 8 m (length) $\times$ 6 m (width) $\times$ 3 m (height), which is the same as that of the neutral context. In particular, we set the color of the neutral context to black to get rid of the possible influence of the environment. Similarly, the two teachers are almost the same except for their facial visualization styles. The virtual teachers repeat an idle standing animation with slight body movements, and show three facial expressions (happy, sad and angry, as shown in Fig. 2) in a continuous loop during the whole experiment. The reason to choose these three expressions is that these are the most common expressions of teachers. The head scales of the two teachers could

^{*}equal contributors

[†]e-mail: han.yu@outlook.com

[‡](✉) e-mail: liuyue@bit.edu.cn

be changed via the press of the up or down button of the Vive controller. Based on the prior work [2], we choose “fixed eye height” as our scaling method, which means that the body scale is adjusted along with the change of head scale so that the virtual teacher always maintains the same eye height. During the experiment, participants are asked to stand still, adjust the head scale of each teacher and select their preferred ideal scale, which means that they could recognize all three facial expressions without feeling uncanny. When they think the head reaches an ideal scale, they need to squeeze the trigger button to confirm the head scale, which would also advance the experiment to the next trial condition. In total, each participant experiences 12 trials. The order of the experiment is balanced using a Latin square design to prevent a potential order effect.

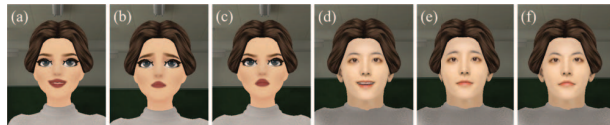


Figure 2: Three facial expressions of virtual teachers with different facial visualization styles: (a) happy, cartoon; (b) sad, cartoon; (c) angry, cartoon; (d) happy, realistic; (e) sad, realistic; (f) angry, realistic.

2.4 Procedure

In the beginning, participants were briefly instructed on the content and procedure of the experiment as well as the use of the Vive controller. After filling out a consent form and a demographic questionnaire, participants wore VR headsets with the help of experimenters. Before the actual experiment, participants have a chance to understand their experimental tasks and the operation of the Vive controller during a practice session. Once participants got familiar with the devices and the VR environment, the actual experiment and data collection started. After completing all the experimental processes, participants took off the VR headsets.

2.5 Data Collection and Analysis

The experimental data (i.e. the head scales) was recorded by the system once the participants squeeze the trigger button of the Vive controller. Data were exported to a Microsoft Excel file after all participants completed the experiment. IBM SPSS Statistics 25 was adopted as the main software for statistical analysis.

3 RESULTS AND DISCUSSION

Within the scope of this study, participants’ preferred ideal head scales for recognizing teachers’ facial expressions were determined. Before statistical analyses, a Shapiro-Wilk test was applied and the results showed that data were not normally distributed. Hence, the non-parametric Scheirer-Ray-Hare test was performed. Data are reported as statistically significant at $p < 0.05$. * and *** correspond to the significance levels of $p < 0.05$ and $p < 0.001$, respectively. The results indicated significant main effects of facial visualization style ($H(1, 227) = 4.404, p = 0.036, \eta^2 = 0.015$) and distance ($H(2, 227) = 122.194, p < 0.001, \eta^2 = 0.534$). Moreover, there was no significant main effect of surrounding context ($H(1, 227) = 0.053, p = 0.818$). No significant interaction effect was found.

For the comparison between different facial visualization styles, the results of the Scheirer-Ray-Hare test showed that the difference of head scales was significant, indicating that the cartoon-styled face leads to smaller head scales, as shown in Fig. 3 (a). Besides, follow-up post-hoc Kruskal-Wallis H tests were conducted to evaluate pairwise differences across the levels of distance. The results showed that all pairwise differences were significant, indicating that the head scales of virtual teachers increased significantly with the increase of the distance, as shown in Fig. 3 (b).

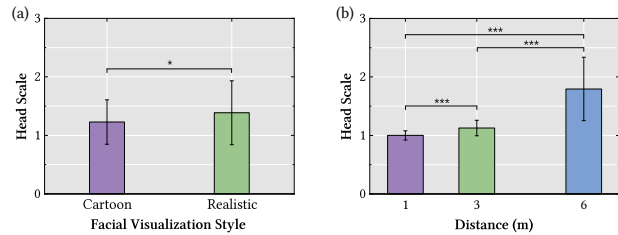


Figure 3: Key results from statistical analyses: (a) comparison between facial visualization styles; (b) comparison between distances.

Our results showed that participants usually increased the head scales of the teachers in order to recognize facial expressions. The results could be attributed to the fact that HMD’s limited screen resolution made it difficult to recognize facial details for longer distances. What’s more, participants’ preferred ideal head scales of the realistic-styled face were generally larger than those of the cartoon-styled face. A possible explanation for this difference may be that the facial features of the cartoon-styled face were a little exaggerated, which made it easier to recognize facial expressions.

4 CONCLUSION

In this paper, we investigated the impacts of design and environmental factors on humans recognizing facial expressions in educational VR through a human-subject study. Our results showed that the facial visualization style of virtual teachers and the distance from the user to the virtual teacher were two factors that affected users’ visual perception in facial-expression-recognizing tasks, whereas the surrounding context had little influence. This research gives evidence that facial visualization styles did influence a user’s perception of facial expressions. The head scales of teachers need to be increased to ensure that users can recognize facial expressions when teachers move away from users. Future work could investigate which of the facial features has the largest influence on the recognition of facial expressions, such as the size and style of eyes, eyebrows, mouth, and so on. The connection between the size/style of facial features and the chosen head scales should be further studied.

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